



Regenerative Horticulture in Fruit Crops: A Sustainable Pathway for Resilient Orchards

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Abstract

Regenerative horticulture is an emerging approach that aims to restore soil health, enhance biodiversity, improve ecosystem resilience and build long-term productivity in fruit orchards. Unlike conventional horticulture that depends heavily on synthetic fertilizers, pesticides and intensive tillage, regenerative systems are based on ecological principles that rebuild soil organic matter, reduce external inputs and improve natural nutrient cycling. This article highlights the principles, practices, crop-specific applications, benefits, challenges and scientific evidence supporting regenerative horticulture in fruit crops.

Introduction

Fruit crops are long-duration, perennial systems that significantly influence soil structure, biodiversity and microclimate. Over the past few decades, the intensive cultivation of fruit orchards-characterized by excessive tillage, monocropping, chemical overuse and low organic matter-has resulted in soil degradation, declining fertility and increased vulnerability to climate stress.

Regenerative horticulture offers an ecologically balanced alternative. Its goal is not only to sustain production but also to restore ecosystem function, improve farmer profitability and ensure resilience to drought, heatwaves, pests and diseases. In fruit crops such as mango, banana, pomegranate, grape, citrus and guava, regenerative practices can significantly improve soil organic carbon, water-use efficiency, fruit quality and yield stability.

Core Principles of Regenerative Horticulture

Regenerative horticulture is based on a set of interlinked ecological principles that enhance the natural functioning of orchard ecosystems.

Building Soil Organic Matter

Increasing soil organic matter (SOM) is central to regenerative systems. SOM improves soil structure, nutrient holding capacity, aeration and water retention.

Practices include: Compost, FYM and vermicompost application, green manuring (sunhemp, cowpea, daincha), use of compost teas and liquid organic manures, regular incorporation of leaf litter and orchard waste and higher SOM promotes microbial diversity, enhancing nutrient mineralization and disease suppression.

Minimizing Soil Disturbance

Frequent ploughing disrupts soil aggregates, kills beneficial organisms and accelerates carbon loss.

Regenerative practices include: No-till or minimum-tillage orchard management, mulch-based weed control instead of deep tillage and planting cover crops rather than exposing bare soil this helps maintain soil structure and underground biodiversity.

Diversification and Multi-Species Integration

Monoculture orchards are prone to pest outbreaks and soil nutrient imbalances. Diversification strategies: Intercropping vegetables, legumes or medicinal crops, multi-species cover crops (legumes + grasses + brassicas), integration of flowering strips to attract pollinators and mixed fruit orchards this increases nutrient cycling, reduces pest pressure and boosts farmer income.

Permanent Soil Cover

Bare soil loses moisture, erodes easily and supports fewer beneficial microbes. Types of soil cover: Organic mulches: dry leaves, straw, grass clippings, living mulches: cowpea, clover, stylosanthes pruned biomass laid on orchard floors and soil cover moderates’ temperature, reduces evaporation and supports moisture retention.

Biological Pest Management

Regenerative horticulture encourages ecological pest control instead of chemical sprays.

Common methods: Use of bioagents (*Trichoderma*, *Pseudomonas*, *Bacillus* spp.) entomopathogenic fungi (*Beauveria*, *Metarhizium*) pheromone, sticky and light traps, botanical extracts (neem oil, garlic-chilli extract) habitat management to encourage predators like ladybirds and lacewings this reduces pesticide residues and enhances orchard biodiversity.

Water Conservation and Regenerative Irrigation

Water is a major limiting factor in fruit production. Regenerative approach emphasizes efficient distribution, storage and soil-based conservation.

Techniques include: Drip and micro-sprinkler irrigation, rainwater harvesting structures: trenches, pits, check dams, mulching to reduce evaporation, contour farming on slopes and improved water retention enhances fruit size, quality and yield stability.

Recycling Orchard Biomass

Fruit orchards produce significant biomass in the form of pruned branches, fallen leaves and rejected fruits.

Regenerative use includes: Composting or vermicomposting orchard waste, biochar production from pruned branches, extracts and teas from compost or fermented materials and returning biomass to soil rather than burning, this closes the nutrient loop and reduces dependence on external fertilizers.

Regenerative Approaches for Major Fruit Crops

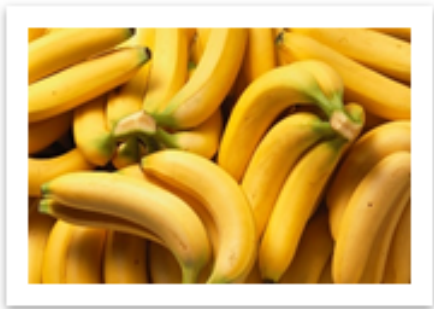
Mango

- Mulching with grasses or leaf litter to reduce alternate bearing stress
- Cover crops such as cowpea or velvet bean
- Biofertilizers (*Azospirillum*, PSB, mycorrhizae)
- Drip irrigation with organic fertigation
- Flowering strips to improve pollination
- **Benefits:** higher fruit set, improved soil moisture, reduced weed growth.



Banana

- Heavy mulching using banana trash and dry leaves
- Intercropping with legumes (groundnut, cowpea)
- Vermicompost + Trichoderma mixture for healthy root zone
- Drip irrigation for uniform bunch development
- **Benefits:** reduced weevil damage, improved bunch weight, enhanced soil carbon.



Citrus

- Living mulches to suppress citrus greening vector movement
- Organic mulches prevent fruit cracking
- Biofertilizer-based nutrition (Azotobacter + VAM)
- **Benefits:** better fruit juice content, greener canopy, reduced soilborne diseases.



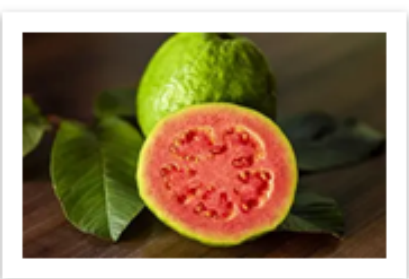
Grapes

- Multi-species cover crops to reduce soil salinity
- Mulching for berry size improvement
- Biological control against thrips and mealybugs
- **Benefits:** reduced chemical sprays, improved TSS and berry uniformity.



Pomegranate, Papaya, Guava

- Mulching + drip reduces cracking and sunburn
- AMF and PGPR for enhanced nutrient uptake
- Habitat strips improve pollination and fruit set
- These systems improve resilience to drought and heatwaves.



Benefits of Regenerative Horticulture in Fruit Orchards

Improved Soil Health

- 20–40% increase in soil organic carbon in 3–5 years
- Higher microbial biomass and enzyme activity
- Improved soil structure and reduced compaction

Enhanced Water-Use Efficiency

- 30–50% water savings under mulching + drip
- Better infiltration and reduced evaporation
- Lower irrigation cost

Increased Biodiversity

- More pollinators, predators and decomposers
- Lower risk of pest outbreaks
- Improved orchard stability

Higher Yield Stability and Fruit Quality

- Better fruit size, shelf-life, TSS, aroma, colour
- Reduced cracking and sunburn incidents
- Resilience to climate stress

Reduced Input Costs

- Lower fertilizer and pesticide costs
- Reduced labour for weeding
- Improved long-term profitability

Challenges in Implementing Regenerative Horticulture

- Initial yield fluctuations during the transition phase
- Need for locally available biomass for mulching
- Limited knowledge among farmers
- Slow nutrient release from organic sources
- Institutional and market support still developing
- Despite challenges, long-term benefits far outweigh short-term limitations.

Conclusion

Regenerative horticulture offers a transformative pathway for sustainable fruit production. By restoring soil health, improving biodiversity, reducing external inputs and enhancing climate resilience, regenerative practices can significantly improve orchard productivity and farmer income. With growing awareness, research support and market demand for eco-friendly fruits, regenerative horticulture is poised to become the foundation of future fruit production systems.

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